

Secretary Problem Research

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Lecture 8: Partially Adversarial SP with Examination

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1 Partially Adversarial Secretary Problem (PASP)

1.1 Problem formulation

Random arrival. n items are totally ordered. A random permutation σ is taken uniformly from the permutation group S_n . The classical SP corresponds to finding a stopping time that most likely stops on $t = \sigma(1)$. We will always use numbers to denote the true rank of items, so the incoming stream of items is represented by $\sigma(1, \dots, n)$.

Adversarial noise. The adversary adds a random noise with resultant permutation π . This permutation is not over the arriving order of object, but over the observed rank, i.e. $\pi(j) = i$ means that the item with actual rank j will be found by the player to be i -th best, if the player observes all items without examining them.

Restriction on adversary. To restrict the power of adversary, we demand π to be long-range stable, i.e. $\pi(i) < \pi(j)$ whenever $i < j$ and $j - i > c$, where the constant c is known to the algorithm.

Examination. For each incoming item the player can choose from three options: Accept, Decline, and Examine. If the player chooses to Examine, the current item will be added to a set E of examined items, and the next item will be forfeited and left unobserved. We use E_t to denote the items examined by the start of t -th round. The Examine option is not available if the current item is the last item.

Observable quantities. Let O_t be the observed total order on round t . It is a total order on $[t]$ induced by $\pi\sigma^{-1}$. Let $\hat{O}_t$ be the actual total order on round t . It is a total order on $[t]$ induced by σ^{-1} . Let P_t be the total order obtained from examination, defined by restricting $\hat{O}_t$ to E_t . Note that O_t and P_t may not be compatible since O_t is subject to observational "mistakes" caused by the noise. We use $\geq_{O_t}$ to denote the order inside the ordered set O_t .

It is equivalent, but sometimes more convenient, to use the notion of relative rank instead of total order. We define $\text{RR}_t(i) = \sum_{s=1}^t \mathbf{1}(\pi\sigma^{-1}(s) \geq \pi\sigma^{-1}(i))$ to be the observed relative rank function.

Decision rule. Let $(\mathcal{F}_t)_{t \in [n]}$ be the sequence of σ -algebras with each $\mathcal{F}_t$ generated by the random variables $O_1 \times P_1, \dots, O_t \times P_t$, that is,

$$\mathcal{F}_t = \sigma(O_1 \times P_1, \dots, O_t \times P_t) = \sigma(O_t \times P_t).$$

By a stopping time, we mean a random variable τ taking values in $[n]$ and satisfying the property $\{\tau = t\} \in \mathcal{F}_t$. The decision rule is a stopping time τ together with a sequence of indicators $\text{Exm}_t := \mathbf{1}(t \in E_{t+1})$ such that

$$\text{Exm}_{t+1} \in \mathcal{F}_t \text{ and } \text{Exm}_t \text{Exm}_{t+1} = 0 \text{ and } \mathbf{1}(\tau = t) \text{Exm}_t = 0.$$

Goal. Our goal is to choose a decision rule (τ, Exm) to maximize $\Pr[\tau = \sigma(1)]$.

1.2 Preliminaries

We list several obvious facts about this problem construction.

- Lemma 1.** 1. The relative ranks behave the same: $RR(t)$ is uniformly sampled from $1, \dots, t+1$ and independent from each other.
2. When π do not depend on σ , $\sigma\pi$ and σ have exactly same distribution.

1.3 Case without examination

We first do not allow examination, and see how far the algorithm can get.

1.3.1 Weak adversarial, do not know anything

Consider the weakest adversary, that can only pick a single π without knowing about the algorithm. The competitive ratio is

$$\begin{aligned} & \inf_{\pi} \sup_{\tau} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma\pi(1)]] \\ &= \inf_k \inf_{\pi: \pi(1)=k} \sup_{\tau} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma(k)]] \\ &\leq \inf_k \sup_{\tau} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma(k)]] . \end{aligned}$$

The problem reduces to a generalized post-doc variant of SP, where we want the item with absolute rank k instead of 1. Each time k is provided, we may specifically design our algorithm to maximize the probability of choosing the rank k item. This is significantly harder to solve as compared to the classical SP, in its value function is no longer monotone: the function

$$\Pr [\text{AR}(t) = m | \text{RR}(t) = r] = \frac{\binom{m-1}{r-1} \binom{n-m}{t-r}}{\binom{n}{t}}$$

is not monotone wrt r for any fixed m . No explicit solution has been found so far. We can show that the worst case when $c < \frac{n}{2}$ is attained at $\pi(1) = c$, and at $\pi(1) = \lfloor \frac{n}{2} \rfloor$ when $c \geq \frac{n}{2}$.

1.3.2 Weak adversarial, know algorithm

Now we assume that the adversary adapts to each algorithm, but cannot depend on the random permutation σ . The competitive ratio is now

$$\sup_{\tau} \inf_{\pi} \mathbb{E}_{\sigma} [\Pr [\tau = \sigma\pi(1)]]$$

Note that the generalized postdoc variant of SP is an upper bound to the problem:

$$\begin{aligned} &= \sup_{\tau} \inf_k \inf_{\pi: \pi(1)=k} \mathbb{E}_{\sigma} \Pr [\tau = \sigma(k)] \\ &\leq \sup_{\tau} \inf_k \mathbb{E}_{\sigma} \Pr [\tau = \sigma(k)] . \end{aligned}$$

This time we need to tackle all sub-problems using only one algorithm. In other words, the algorithm's probability to choose the 1, 2, ..., c -th item are compared, and the minimum one is the CR.

For lower bound, we consider the following algorithm: When each item arrives, if $RR(t) > c$, reject the item. Otherwise, accept with probability p , where p is a tuning parameter. Now for some algorithm τ and fixed k , We derive analytic result for this algorithm. Its performance monotonously decrease wrt $\pi(1)$ for fixed n and p , and always approach the trivial lower bound $\frac{1}{n}$.

Consider another algorithm: Let the first pn items to go by, and then select the first item with relative rank better or equal to c . This algorithm makes no distinction among the first c items, so it suffices to analyze its behavior on the classic SP.

1.3.3 Strong adversarial, knows τ and σ

$$\sup_{\tau} \mathbb{E}_{\sigma} \inf_{\pi} [\Pr [\tau = \sigma\pi(1)]] .$$

1.3.4 Strong adversarial, knows τ , σ and $\mathbf{r}$

$$\sup_{\tau} \mathbb{E}_{\sigma} \left[\mathbb{E}_{\mathbf{r}} \left[\inf_{\pi} \mathbf{1} (\tau = \sigma\pi(1)) \right] \right] .$$

In this case the adversary knows the random seed used by the algorithm, so it's able to make hindsight decisions. In this setting, the algorithm achieve zero cr. This can be shown by calling to an auxiliary problem of adversarial best-choice within the top c items, where no deterministic algorithm can guarantee nonzero competitive ratio against an adversary.

1.4 Lower Bound for an algorithm

We establish a lower bound for the following simple algorithm: examine each item with $RR_t(t) \leq c$. Accept the current item if $t \geq T$ and the item is the best so far.

Algorithm 1 Simple algorithm

```

1: while At least two items are left do
2:    $x \leftarrow$  the current item to be decided
3:    $t, O_t, P_t \leftarrow$  as defined before
4:   if  $x \in P_t$  then
5:     if  $x = \max P_t$  and  $t \geq T$  then
6:       Accept  $x$ 
7:     else
8:       Decline  $x$ 
9:     end if
10:  else if  $|\{a \in O_t : a \geq_{O_t} x\}| \leq c$  then  $\triangleright$  Means  $RR_t(x) \leq c$ 
11:    Examine  $x$ 
12:  else
13:    Decline  $x$ 
14:  end if
15: end while
16: Accept the only item left

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Lemma 2 (Equivalent construction of the game). It is equivalent to view this game as follows: when an item is examined, instead of omitting the next item, the last available item is taken from the end of the queue. The player wins only if the best item is accepted and is not omitted.

Proof. Consider a deterministic algorithm ALG . The algorithm never knows $\sigma^{-1}(t)$, the position of the current item in the arrival sequence. For some $\sigma \in S_n$, we can define an automorphism $\varphi(\text{ALG}) \in A(S_n)$ according to the following reorder of elements: We first let the algorithm run through the sequence, mark all the items omitted, and move the marked items to the end of the queue. The resultant sequence σ' looks exactly the same to the algorithm in the new setting.

In the original setting we have the sequence of incoming items

$$\sigma^{-1}(t) \quad \sigma^{-1}(t+1) \quad \sigma^{-1}(t+2) \quad \dots \quad \sigma^{-1}(p).$$

When ALG encounters $\sigma^{-1}(t)$ and choose to **Examine**, $\sigma^{-1}(t+1)$ will be omitted and $\sigma^{-1}(t+2)$ is revealed to ALG .

On the other hand, in the modified setting, after applying $\varphi(\text{ALG})$ we have the incoming sequence

$$\sigma^{-1}(t) \quad \sigma^{-1}(t+2) \quad \dots \quad \sigma^{-1}(p) \quad \sigma^{-1}(t+1).$$

examining $\sigma^{-1}(t)$ will drop the last item in the sequence, while $\sigma^{-1}(t+2)$ is observed next, identical to the original case.

We have shown that the original and modified setups of the game are equivalent. $\square$

Intuition. Imagine a strip of paper which represents the incoming sequence of items. One consumes the items from left to right, but the paper strip burns from right to left (due to examination which makes some items unavailable). The moment the paper is used up, the game is forced to end.

Let $t^* := \sigma^{-1}(1)$, i.e., the position of the best item. We couple the game with the classical secretary problem (CSP).

Lemma 3 (Coupling with CSP). Consider any deterministic algorithm ALG . Then ALG wins $\text{PASP}(n, c)$ implies ALG wins $\text{CSP}(n)$. Conversely, ALG wins $\text{CSP}(n)$ implies ALG wins $\text{PASP}(n, c)$ whenever $t^* + |E_{t^*}| \leq n$.

Proof. Assume an algorithm ALG wins $\text{PASP}(n, c)$. Then we can use Lemma 2 to move all omitted items to the end of the queue, so the algorithm does not accept any item before 1. Now the same algorithm, in the CSP setting, observe all items before 1 and accepts 1.

Assume the algorithm wins $\text{CSP}(n)$. We can perform the process above in the opposite way: for each item that ALG decides to examine, pull an element from end of the queue and put it right next to the examined item. The inequality

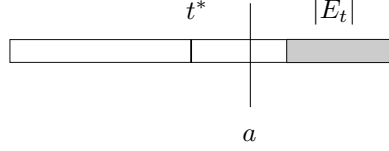


Figure 1: Our scheme to finding a lower bound

$t^* + |E_{t^*}| \leq n$ guarantees that there are enough elements at the end of the queue to proceed with our construction.

□

Lemma 4. For any a and $n > a$,

$$\begin{aligned} \Pr[\text{win CSP with } a \text{ items } | t^* + |E_{t^*}| \leq n] \\ \geq \Pr[\text{win CSP with } a \text{ items}] \end{aligned}$$

Proof. From Lemma 2, we conclude that

$$\text{ALG wins} \Rightarrow t^* + |E_{t^*}| \leq n. \quad (1)$$

□

Theorem 1. (Lower Bound) For PASP with parameters (n, c) , for all $a \in \{1, \dots, n\}$ and $\beta > 0$ such that

$$c \left(\beta + \sum_{s=c}^a \frac{1}{s} \right) + c - 1 \leq n - a$$

we have

$$\Pr[\text{ALG win}] \geq \frac{1}{e} \frac{a}{n} \left(1 - \exp \left(-\frac{2\beta^2}{a - c + 1} \right) \right).$$

Proof. From Lemma 3 we have

$$\Pr[\text{win}] = \Pr[\text{win CSP with } n \text{ items and } t^* + |E_{t^*}| \leq n] \quad (2)$$

Now we constrain t^* and $|E_{t^*}|$ by a to make sure they don't meet.

$$\geq \max_{a \leq n} \Pr [\text{win CSP with } n \text{ items and } t^* + |E_{t^*}| \leq n \cap t^* \leq a] \quad (3)$$

Which follows from property of intersection. Continue to apply elementary properties,

$$= \max_{a \leq n} \Pr [\text{win CSP with } n \text{ items } |t^* + |E_{t^*}| \leq n \cap t^* \leq a] \quad (4)$$

$$\Pr [t^* + |E_{t^*}| \leq n | t^* \leq a] \Pr [t^* \leq a]$$

$$= \max_{a \leq n} \Pr [\text{win CSP with } n \text{ items } |t^* + |E_{t^*}| \leq n \cap t^* \leq a] \quad (5)$$

$$\Pr [|E_{t^*}| \leq n - a] \Pr [t^* \leq a]$$

Since a CSP with n items where only the first a items may be the best is identical to a CSP with a items,

$$= \max_{a \leq n} \Pr [\text{win CSP with } a \text{ items } |t^* + |E_{t^*}| \leq n] \quad (6)$$

$$\Pr [|E_{t^*}| \leq n - a] \Pr [t^* \leq a]$$

By Lemma 4

$$\geq \max_{a \leq n} \Pr [\text{win CSP with } a \text{ items}] \quad (7)$$

$$\Pr [|E_{t^*}| \leq n - a] \Pr [t^* \leq a]$$

Evaluate each term separately,

$$\Pr [\text{win}] \geq \max_{a \leq n} \frac{1}{e} \Pr [|E_a| \leq n - a] \Pr [t^* \leq a]. \quad (8)$$

For $t \geq c$ we have

$$|E_t| = c - 1 + c \sum_{s=c}^t B_s \quad (9)$$

where $B_s \sim \text{Ber}(\frac{1}{s})$.

Using Hoeffding's Bound:

$$\Pr \left[\sum_c^t B_s - \sum_c^t \frac{1}{s} \leq \beta \right] \geq 1 - \exp \left(-\frac{2\beta^2}{t - c + 1} \right) \quad (10)$$

$$\Pr \left[|E_t| \leq c(\beta + \sum_{s=c}^t \frac{1}{s}) + c - 1 \right] \geq 1 - \exp \left(-\frac{2\beta^2}{t - c + 1} \right). \quad (11)$$

Fix some a . Let β be the greatest real number such that

$$c(\beta + \sum_{s=c}^a \frac{1}{s}) + c - 1 \leq n - a. \quad (12)$$

Therefore,

$$\beta = \frac{n - a}{c} - 1 - H(a) + H(c - 1) \quad (13)$$

where we use $H(a)$ to denote the a -th harmonic number.

Now we have a concentration bound for $|E_{t^*}|$:

$$\Pr[|E_{t^*}| \leq n - a] \geq 1 - \exp\left(-\frac{2\beta^2}{t - c + 1}\right). \quad (14)$$

Applying this bound, we now have

$$\Pr[\text{win}] \geq \max_{a \leq n} \frac{1}{e} \frac{a}{n} \left(1 - \exp\left(-\frac{2\beta^2}{a - c + 1}\right)\right). \quad (15)$$

□

Numerical Result. Take $c = 2$ and $n = 100$ for an example. When $a = 70$, we have the numerical bound $\Pr[\text{win}] \geq 0.674 \frac{1}{e}$, which is a non-trivial improvement from the examine-everything algorithm which guarantees only a competitive ratio of $0.5 \frac{1}{e}$.

Take $c = 1$ and $n = 100$, then the optimal choice of a is 82, and the corresponding bound is $\Pr[\text{win}] \geq 0.807 \frac{1}{e}$.

Lecture 9: Upper Bound

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1.5 Upper Bound

Theorem 2. When $c = 1$, there exists a family of problem instances where the best CR possible is $\frac{1}{2e}$.

Proof. For any outcome $\omega \in \Omega$, the problem instance is uniquely identified with $\sigma(\omega)$. We consider all problem instances where the best and second best items are next to each other. More precisely, we restrict the probability space to the following event:

$$|\sigma(\omega)(1) - \sigma(\omega)(2)| = 1.$$

Which in term gives the restricted range for σ :

$$\mathcal{S} = \{\sigma \in S_n : |\sigma(1) - \sigma(2)| = 1\}.$$

As discussed before, in the restricted probability space, items 1 and 2 are not distinguishable by any algorithm. Therefore, we can treat these two items as a unit, and construct the coupling problem instance below:

Let σ' be a random permutation taking value in S_{n-1} . We additionally require

$$\sigma'(i) < \sigma'(j) \Leftrightarrow \sigma(i+1) < \sigma(j+1)$$

for all $i, j \in \{2, 3, \dots, n-1\}$. We may verify that this uniquely defines σ' in terms of σ , and σ' is uniformly distributed in S_{n-1} .

Now consider the game instance with $n-1$ items and $c = 0$, identified by σ' . This coupling game is same as a classical SP with $n-1$ items, where any algorithm has $\text{CR} \leq \frac{1}{e}$. Also, for any fixed algorithm τ with σ as an input, winning outcomes in the two coupling games are related by

$$\tau(\sigma) = \sigma(1) \vee \tau(\sigma) = \sigma(2) \Leftrightarrow \tau(\sigma') = \sigma'(1).$$

Since any algorithm cannot make a distinction between the events $\tau = \sigma(1)$ and $\tau = \sigma(2)$,

$$\Pr[\tau(\sigma) = \sigma(1)] = \Pr[\tau(\sigma) = \sigma(2)].$$

To justify this, note that for any problem instance $\sigma \in \mathcal{S}$, there exists $\bar{\sigma} \in \mathcal{S}$ where $\bar{\sigma}(1) = \sigma(2)$ and $\bar{\sigma}(2) = \sigma(1)$. We must have $\tau(\sigma) = \tau(\bar{\sigma})$, so τ makes the correct decision half of the time on average.

Hence we have the probability of winning bounded by

$$\Pr[\tau = \sigma(1)] = \frac{1}{2} \Pr[\tau' = \sigma'(1)] \leq \frac{1}{2e}.$$

□